

# RIDMIKA YOHANI EKANAYAKA

Undergraduate

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## PERSONAL SUMMARY

Final-year Software Engineering undergraduate with a strong interest in Data Analytics and hands-on experience in data analysis, visualization, and software development. Skilled in Python, SQL, Power BI, and data visualization, with a good understanding of data cleaning, exploratory data analysis, and statistical analysis. Seeking an internship opportunity to apply my technical and analytical skills while gaining practical industry experience.

## EDUCATIONAL QUALIFICATIONS

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|-----------------------------------------------------------------------------------|----------------|
| <b>BSc (Hons) in Software Engineering</b> , Sri Lanka Technology Campus, Padukka  | 2023 – Present |
| <b>G.C.E. Advanced Level (Physical Science)</b> , Ferguson High School, Ratnapura | 2021           |
| <b>G.C.E. Ordinary Level</b> , Ferguson High School, Ratnapura                    | 2018           |
| – Results: A – 5, B – 2, C – 2                                                    |                |

## TECHNICAL SKILLS

**Data Analysis:** Data cleaning, Data interpretation, Statistical analysis, Dashboard Reporting, Power BI Dashboard, Excel

**Tools & Platforms:** Excel, Power BI, Google Colab, MySQL, Jupyter Notebook, Git, GitHub, VS Code, Overleaf, Postman

**Programming:** Python (Pandas, NumPy), SQL, Java, JavaScript, C, C++, HTML, CSS, SCSS

**Libraries & Frameworks:** React, Node.js, Express.js, Flutter, Bootstrap

**Databases:** MySQL, MongoDB

**Data Visualization:** Power BI dashboards, Excel dashboards, chart and report generation, Google Colab visualizations

## PROJECTS

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|------------------------------------------------------------------|------------------------------------|
| <b>Human-Centered Explainable AI Framework for Loan Approval</b> | Final Year Project, Ongoing (2026) |
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AI-driven loan approval systems often act as “black boxes,” making decisions without explaining the reasoning behind them and sometimes treating applicant groups unfairly. This ongoing project proposes an XAI-based framework using tree-based machine learning models and SHAP explanations to make loan approval decisions more transparent, along with fairness metrics to help reduce bias. The system is being developed with a Flutter frontend and a Node.js/Express API backend.

**Technologies:** Python, Machine Learning, SHAP, Fairness Metrics, Flutter, Node.js, Express.js

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| <b>SmartCare – Patient Readmission Prediction</b>   <a href="#">Link</a> | Academic Project, September 2026 |
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Built a machine learning pipeline to predict 30-day hospital patient readmissions using Random Forest, Decision Tree, XGBoost, and KNN models on a 1,000-record clinical dataset. Applied SHAP to provide model explainability and deployed a Streamlit prototype for clinical decision support. Achieved a ROC-AUC score of 0.942 and 100% recall using the Random Forest model.

**Technologies:** Python, Scikit-learn, XGBoost, SHAP, Pandas, Streamlit

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| <b>OGBN-Arxiv GNN Node Classification</b>   <a href="#">Link</a> | Academic Project, August 2026 |
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Implemented and compared Graph Convolutional Network (GCN) and Graph Attention Network (GAT) models for node classification on a 169,343-node citation dataset. Achieved 54.6% test accuracy, analyzed model behavior through embedding visualizations and attention-weight analysis, and presented findings using an interactive Streamlit dashboard.

**Technologies:** Python, PyTorch, PyTorch Geometric, Scikit-learn, Streamlit, Google Colab

## Sri Lankan Retail Price Intelligence Dashboard | [Link](#)

Personal Project, July 2026

Developed an interactive dashboard analyzing 22,449 weekly Sri Lankan retail-price observations from 2020–2024, using Python, Pandas, Plotly, and Streamlit. Built a data-validation and cleaning pipeline to handle custom weekly date formats, missing values, price changes, and volatility metrics. Implemented interactive item filtering, trend analysis, price comparisons, and downloadable data views, with safe handling for empty-state scenarios. Added automated tests using pytest, data-quality reports, methodology documentation, and transparent notes on the limitations of unsupported market and sales metrics.

**Technologies:** Python, Pandas, Plotly, Streamlit, pytest, Excel

## TrendVerse – Fashion E-Commerce Website | [Link](#)

Personal Project, February 2026

TrendVerse is a responsive fashion e-commerce website built with a luxury-style UI and reusable components to give shoppers a premium, visually polished browsing experience across devices. The frontend was developed using React, JavaScript, HTML, and SCSS, focusing on clean layout, consistent styling, and reusable UI components such as product cards, navigation menus, and filters. On the backend, Node.js and Express.js handle core store logic, while MongoDB stores product and application data, giving the site a solid full-stack foundation for browsing and managing fashion items.

**Technologies:** React, JavaScript, HTML, SCSS, Node.js, Express.js, MongoDB

## Utopia – Digital Citizen & Government Services Platform | [Link](#)

Academic Project, July 2025

Utopia is a digital citizen and government services platform designed to simplify how citizens interact with government processes online, reducing reliance on manual paperwork and in-person visits. The backend service was implemented using Node.js and Express, with MongoDB (via Mongoose) used for structured data storage of citizen and government records. JWT (JSON Web Tokens) was used to implement secure authentication and protect user sessions, Multer was used to handle file uploads such as documents and identification records, and Nodemailer was integrated to support automated email notifications, such as confirmations or status updates, as part of the citizen-facing workflow.

**Technologies:** Node.js, Express.js, MongoDB, Mongoose, JWT, Multer, Nodemailer

## CERTIFICATIONS

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**Power BI for Beginners** – Simplilearn

**Quality Assurance Courses with Certificate** – Simplilearn

**Full Stack Development Certificate Course**, AIO-IT SOLUTIONS (PVT) LTD

**Full Stack Development Certificate Course**, SKYRE

**Python (Basic) Certificate** – HackerRank

**Power BI Data Modeling with DAX**, Simplilearn

## VOLUNTEERING & EXTRA-CURRICULAR ACTIVITIES

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**Treasurer**, Computer Society Student Branch Chapter of SLTC IEEE

2025 – 2026

- Managed tentative budgets and final financial reports for student events.
- Supported activities including Codemania v6.0, GitGenius 2026, and CloudSpace v1.0.

**IEEE Member & Volunteer**, SLTC IEEE

2024–2026

- Logistics Sub-Committee Deputy Head, IEEE CS Student Branch Chapter, SLTC — 2024–2025
- Promotion Team Head, IEEE Codemania V5.0 — IEEE CS Student Branch Chapter, SLTC
- Coordination Team Deputy Head, IEEE Day 2025 — IEEE CS Student Branch Chapter, SLTC
- Program Team Deputy Head, Arduino Challenge 2025 — IEEE CS Student Branch Chapter, SLTC
- Program Team Member, Decode Xtreme 2024 / Git Genius 2024 — IEEE CS Student Branch Chapter, SLTC
- Volunteered at Codemania v5.0, GitGenius 2024, and DecodeXtream.
- Served in roles including Head, Deputy Head, and Member for various event activities.